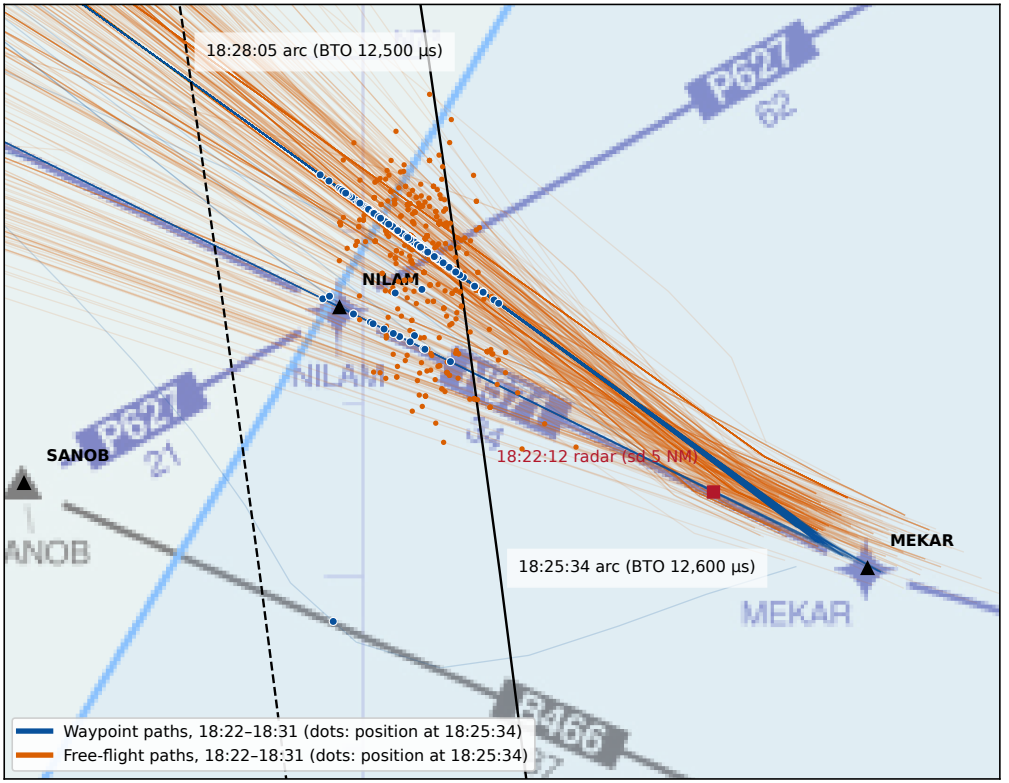
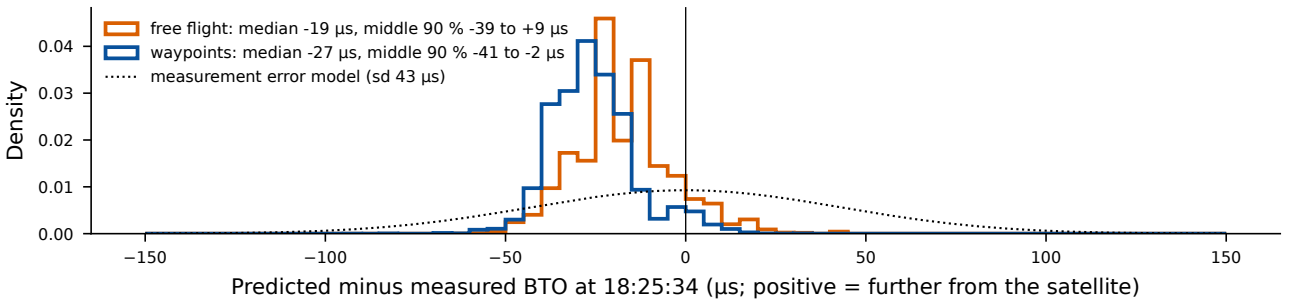
At the 18:25 log-on most waypoint paths are a little past the timing arc; free-flight paths spread across it



Fit to the 18:25:34 timing measurement, all 16,000 posterior draws per family



Simple version (ASD-STE100): The top map shows possible flight paths from 18:22 to 18:31 UTC near MEKAR and NILAM. The dots show where each path is at 18:25:34, when the satellite system logged on again. The black lines show where the timing measurements put the aircraft. The bottom chart shows the difference between each path and the 18:25:34 measurement. The waypoint paths agree a little less well at 18:25:34 than the free paths, because they are all on one line and most of them are a little too far west at that time. This costs the waypoint paths 0.3 of evidence. They gain much more later.

Technical version: Core run I, 8 seeds × 2,000 posterior route draws per family; 40 s trace from 18:22:00, linearly interpolated to 18:25:34; 60 draws per seed shown on the map. BTO predicted with report/satcom_model.py (Inmarsat ephemeris, terminal altitude from the trace) against the logged 12,600 µs; sd 43 µs at 18:25:34 (log-on, BFO excluded), 29 µs at 18:28. Arcs drawn from the engine's reference_arcs (run.json). Evidence increments at 18:25:34 (waypoints minus free) −0.27 nats; 18:22:12 radar −0.16; 18:28 −0.10 and +0.14. Chart background supplied by Pete, affine fit to 12 fixes.